

BPL_TEST2_Fedbatch - demo

```
In [26]: # Import the application model and context variables
from BPL_TEST2_Fedbatch_pyfmi import*
```

```
In [27]: # Import the general FMU_explore functions and adapt them to the application
from fmu_explore_pyfmi import AdaptWith
```

```
Adapted = AdaptWith(parValue, parLocation, parCheck, \
                      model, fmu_model, fmu_process_diagram, \
                      MSL_usage, MSL_version, BPL_version, \
                      opts_std, simulationTime, timeDiscreteStates, \
                      diagrams, ax, lines)

par = Adapted.par
init = Adapted.init
disp = Adapted.disp
simu = Adapted.simu
show = Adapted.show

resetPen = Adapted.resetPen
process_diagram = Adapted.process_diagram
describe_general = Adapted.describe_general
describe_parts = Adapted.describe_parts
describe_MSL = Adapted.describe_MSL
FMU_explore_info = Adapted.FMU_explore_info
system_info = Adapted.system_info
SDG = Adapted.SDG
```

Bring in the user defined FMU_explore functions describe() and newplot()

```
In [28]: run -i BPL_TEST2_Fedbatch_explore.py
```

Model for the process has been setup. Key commands:

- par() - change of parameters and initial values
- init() - change initial values only
- simu() - simulate and plot
- newplot() - make a new plot
- show() - show plot from previous simulation
- disp() - display parameters and initial values from the last simulation
- describe() - describe culture, broth, parameters, variables with values/units

Note that both disp() and describe() takes values from the last simulation and the command process_diagram() brings up the main configuration

Brief information about a command by help(), eg help(simu)

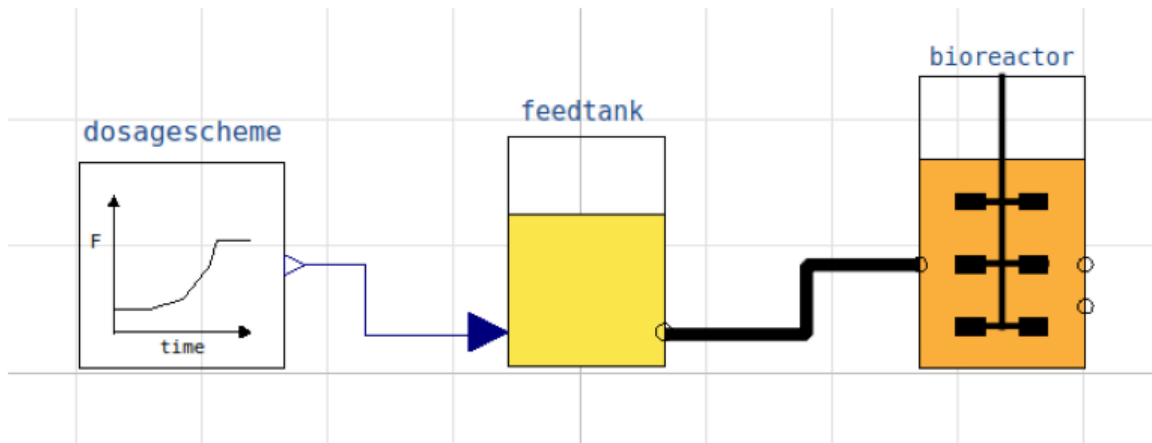
Key system information is listed with the command system_info()

<Figure size 984.252x787.402 with 0 Axes>

```
In [29]: %matplotlib inline
plt.rcParams['figure.figsize'] = [25/2.54, 20/2.54]
```

```
In [30]: process_diagram()
```

No processDiagram.png file in the FMU, but try the file on disk.



```
In [31]: describe('process')
```

```
In [32]: describe('culture')
```

Simplified text book model - only substrate S and cell concentration X

```
In [33]: disp('dosagescheme')
```

```
F_start : 0.0
mu_feed : 0.15
t_startExp : 3.0
F_startExp : 0.006
F_max : 0.3
```

```
In [34]: disp('feedtank')
```

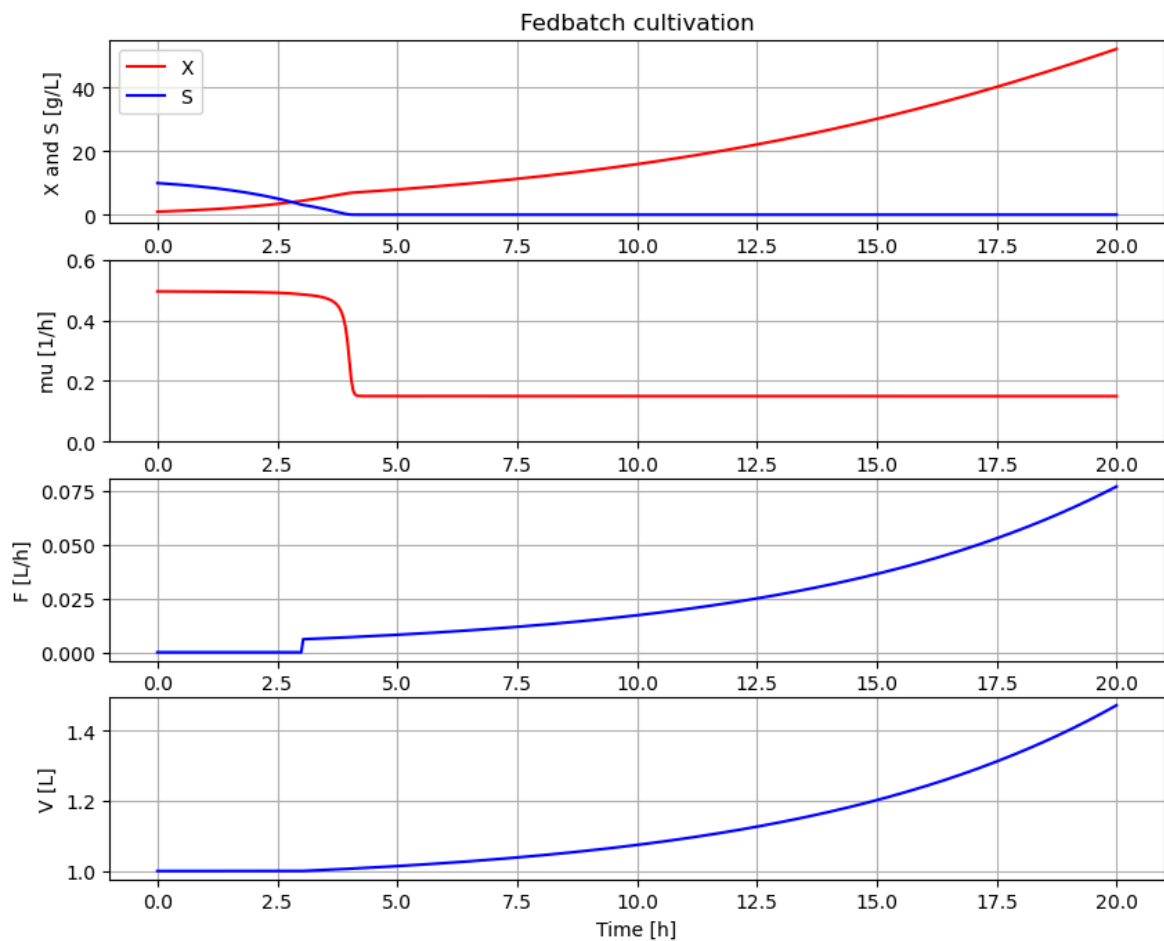
```
feedtank.S_in : 300.0
feedtank.V_start : 10.0
```

```
In [35]: disp('bioreactor', mode='long')
```

```
bioreactor.V_start : V_start : 1.0
bioreactor.m_start[1] : VX_start : 1.0
bioreactor.m_start[2] : VS_start : 10.0
bioreactor.culture.Y : Y : 0.5
bioreactor.culture.qSmax : qSmax : 1.0
bioreactor.culture.Ks : Ks : 0.1
```

```
In [36]: # Simulation with default values of the process
newplot(plotType='TimeSeries'); ax[1].set_ylim(0,0.6)

simu(20)
```

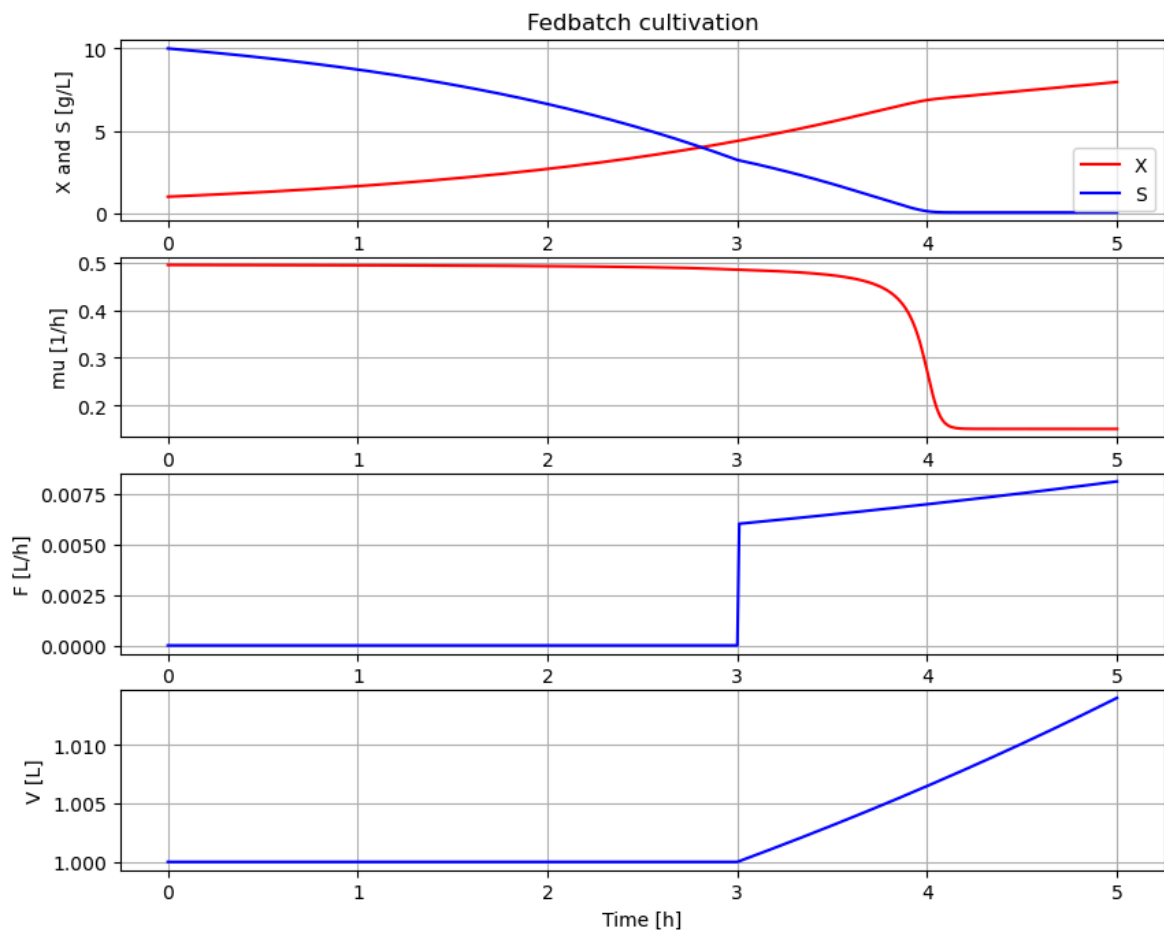


```
In [37]: show()
```

```
In [38]: simu()
```

```
In [39]: show()
```

```
In [40]: newplot()  
show()
```



```
In [41]: # Let us display and then save the feedprofile
         disp('dosagescheme')
```

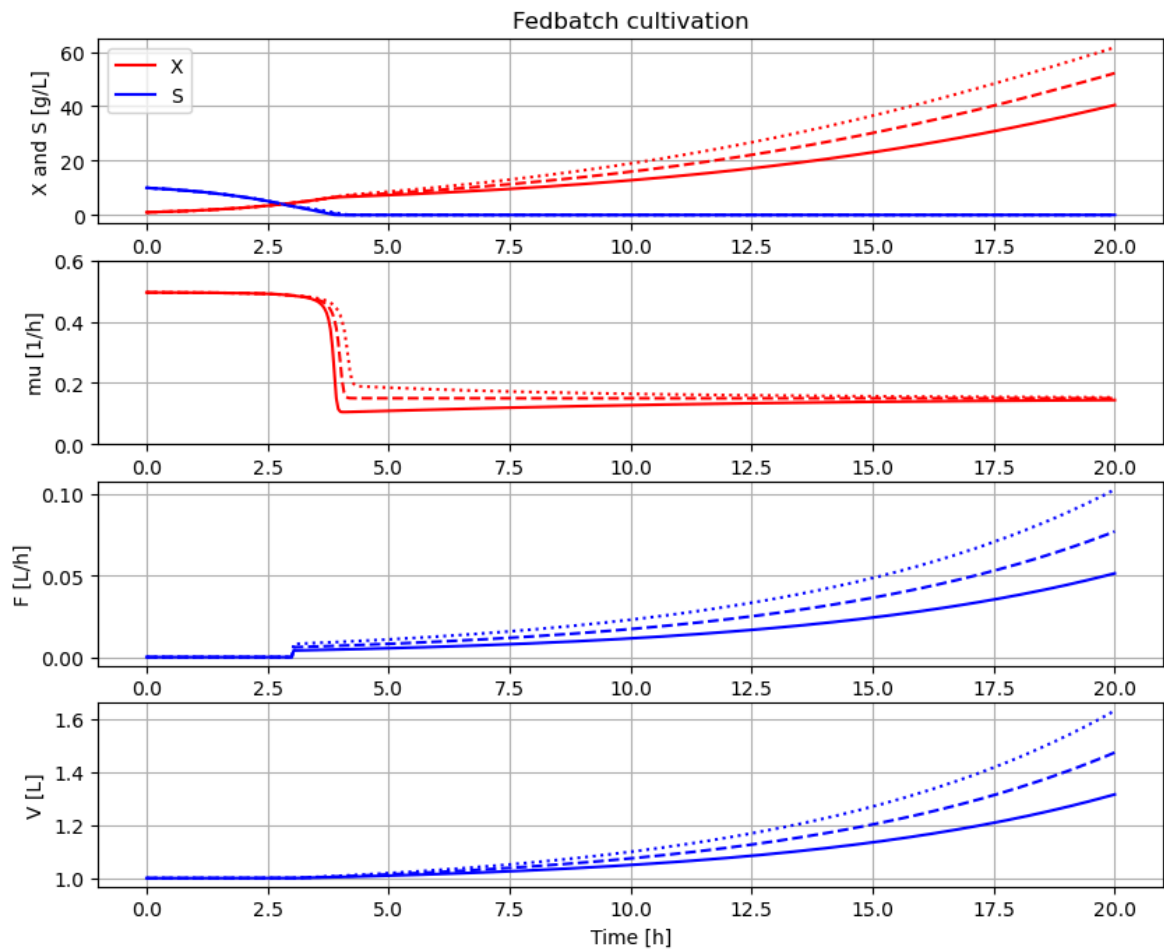
```
F_start : 0.0
mu_feed : 0.15
t_startExp : 3.0
F_startExp : 0.006
F_max : 0.3
```

```
In [42]: feedprofile_1={'F_start':0, 'mu_feed':0.1, 't_startExp':3, 'F_startExp':0.001, 'F_max':0.3}
```

```
In [43]: # Let us develop a feedprofile with mu_feed=0.15 1/h instead
         newplot(plotType='TimeSeries'); ax[1].set_ylim(0,0.6)

         # Choose mu_feed
         par(mu_feed=0.15)

         # Tune F_startExp value to get a stable culture growth rate at mu 0.15 1/h
         for value in [0.004, 0.006, 0.008]:
             par(F_startExp=value)
             simu(20)
```

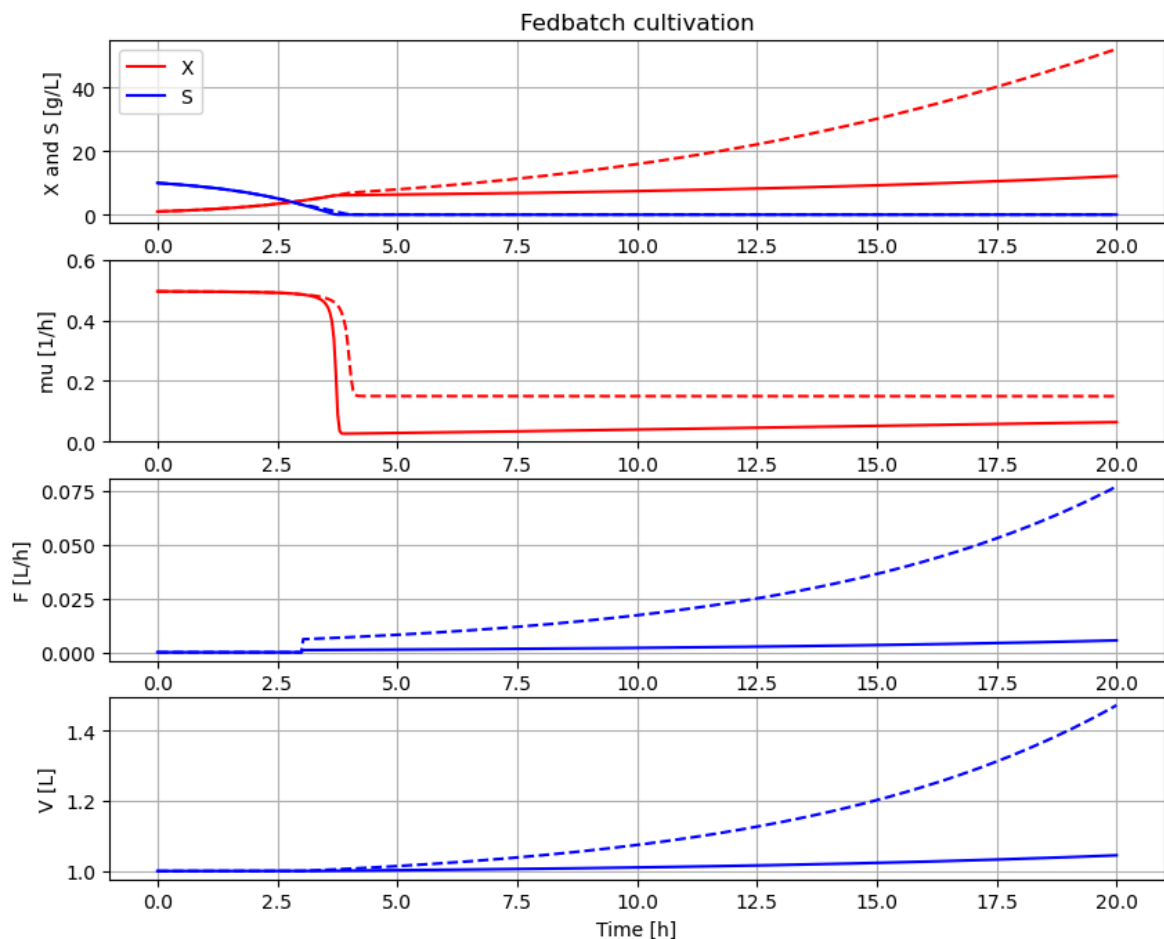


```
In [44]: # We choose F_startExp = 0.008 and save the new feedprofile
feedprofile_2={'F_start':0, 'mu_feed':0.15, 't_startExp':3, 'F_startExp':0.006,
```

```
In [45]: # Compare the result of ot the two
newplot(plotType='TimeSeries'); ax[1].set_ylim(0,0.6)

for feedprofile in [feedprofile_1, feedprofile_2]:
    par(feedprofile)
    simu(20)

# Restore the original feedprofile
par(feedprofile_2)
```



In [46]: `describe('mu')`

Cell specific growth rate variable : 0.15 [1/h]

In [47]: `describe('parts')`

['bioreactor', 'bioreactor.culture', 'dosagescheme', 'feedtank', 'MSL']

In [48]: `describe('MSL')`

MSL: RealInput, RealOutput

In [49]: `system_info()`

System information

- OS: Windows
- Python: 3.12.11
- Scipy: 1.16.0
- PyFMI: 2.22.0
- FMU by: JModelica.org
- FMI: 2.0
- Type: FMUModelCS2
- Name: BPL.Examples_TEST2.Fedbatch
- Generated: 2025-07-26T09:38:43
- MSL: 3.2.2 build 3
- Description: Bioprocess Library version 2.3.1
- Interaction: FMU_explore version 1.2.2

In [50]: `SDG()`